IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE/EIE PART I: MEng, BEng and ACGI

Corrected Copy

DIGITAL ELECTRONICS 1

Friday, 14 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 each carry 30%.

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : Z. Durrani
Second Marker(s) : C. Bouganis

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Special instructions for invigilators: None

Information for candidates:

In the figures showing digital circuits, all components have, unless explicitly indicated otherwise, been drawn with their inputs on the left and their outputs on the right. All signals labelled with the same name are connected together. All circuits use positive logic. The least-significant bit of a bus signal is labelled as bit 0, and the most-significant bit with the highest integer number. Therefore the signal X[7:0] is an eight-bit bus with X7 being the MSB and X0 the LSB.

In questions involving circuit design, you may use any standard digital circuits not explicitly forbidden by the question, provided that you specify fully their operation.

Marks may be deducted for unnecessarily complex designs unless you are explicitly instructed not to simplify your solution.

Question 1

1. a) Simplify the following Boolean expressions using De Morgan's theorem and/or Boolean algebra.

i) $\overline{\overline{A}+\overline{(A\overline{B})}\overline{C}}$

[4]

ii) $ABC+B(A\oplus B)$

[4]

b) Simplify the following Boolean equation, in sum-of-products form, using a Karnaugh map.

$$f=\overline{A}\overline{B}\left(\overline{C\oplus D}\right)+A\overline{B}\overline{C}\overline{D}+ACD+ABD$$

[4]

c) Simplify the following Boolean equation, in product-of-sums form, using a Karnaugh map.

$$f(A,B,C,D)=\sum(2,4,6,8,10,13,14)$$

[4]

d) Assuming that all numbers are 16 bits wide, with binary numbers in 2's complement form, complete the missing entries which are not shaded in the following table. (No marks will be awarded for this question unless you show how the solution is derived.)

[8]

Decimal	Hexadecimal	Binary	BCD
5153	?		
		0011 0111 1001 1000	?
	1CF	?	
-1101		?	

The question continues on the next page

e) The waveforms for signals P and Q shown in Figure 1.1 are applied to the circuit shown in Figure 1.2. Copy the timing diagram and add waveforms for S, T, U and V. Assume that initially U=0 and V=1. You may ignore propagation delays.

[6]

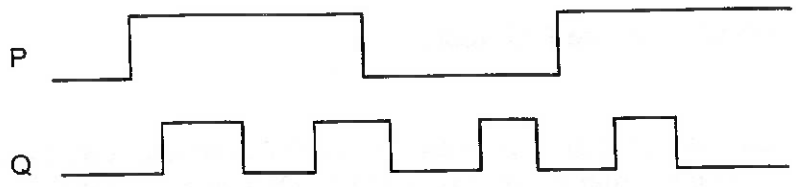


Figure 1.1

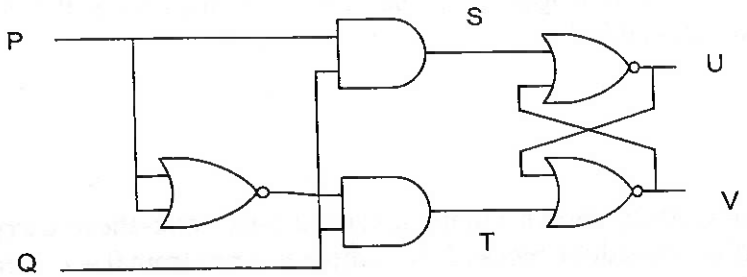


Figure 1.2

f) Sketch the state diagram for the circuit shown in Figure 1.3, where N[2:0] = 011. You may assume that initially, Q[2:0] = 000.

[4]

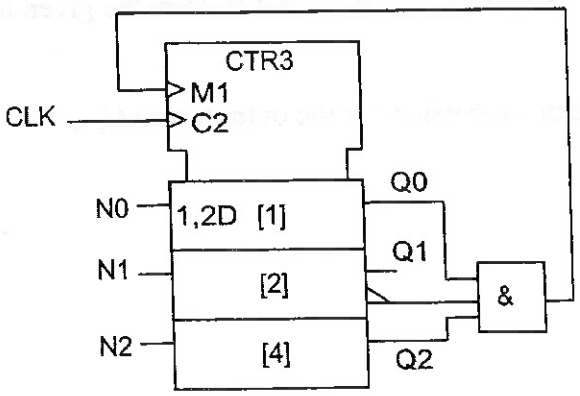


Figure 1.3

g) Implement the function *g* given below using a multiplexer.

$$g(X,Y,Z) = \sum(0,1,2,4)$$

[6]

Question 2

2. a) Figure 2.1 (see next page) shows a Full Adder (FA), with input bits A_0, B_0 and input carry C_0.

(i) Write the truth table for the Full Adder.

[4]

(ii) Hence, by using Boolean algebra or otherwise, find expressions for the outputs S_0 and C_1. Your implementation may use OR, AND, or XOR gates.

[4]

(iii) For a 3-bit ripple adder based on the FA of Fig. 2.1, assuming that each of the constituent gates has a propagation delay time of 10 ns, what is the worst-case delay time for this adder?

[4]

b) Figure 2.2 (see next page) shows a more advanced 3-bit 'look-ahead carry' adder, built using modified full-adder blocks FA_n', where n varies from 0 – 2. Here, rather than the full-adder output carry, signals G_n and P_n are generated instead, and feed into a 'look-ahead carry' (LAC) logic circuit. G_n and P_n are then used to generate the output carry C_{n+1} which feeds into the next full adder block, FA_{n+1}'. G_n is asserted when the input bits A_n, B_n are by themselves sufficient to generate the output carry C_{n+1}. P_n is asserted when a 'high' input carry C_n is necessary to generate C_{n+1}.

(i) Write down Boolean expressions for G_n and P_n from the given logic.

[2]

(ii) Write down a Boolean expression for the output carry C_{n+1}

[2]

The question continues on the next page

(iii) Hence, by writing down Boolean expressions for each output carry C_{n+1}, find the Boolean expression for the final output carry C_3. Assuming this expression is implemented using gates similar to those in part (a), what is the worst-case delay time for this adder?

[4]

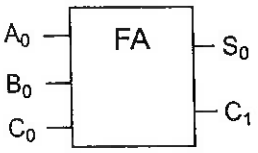


Figure 2.1

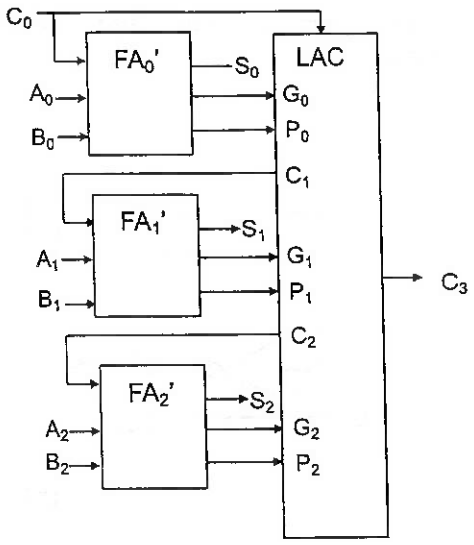


Figure 2.2

The question continues on the next page

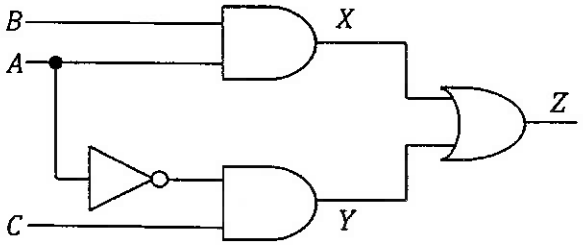
c) For the circuit shown in Fig. 2.3(a), the timing diagram for the input signals A, B and C is shown in Fig. 2.3(b).

(i) Sketch the timing diagram for signals X, Y and Z. Your diagram should include any glitches due to circuit propagation delays.

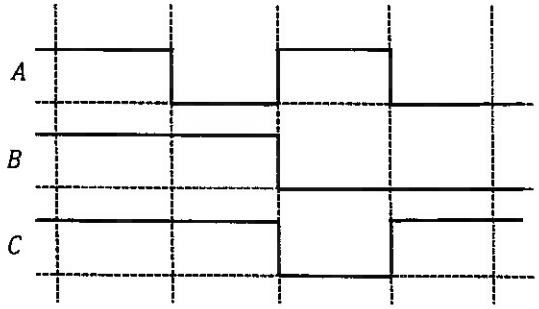
[6]

(ii) Redesign the circuit of Fig. 2.3(a) such that static '1' hazards are prevented from occurring.

[4]



(a)



(b)

Figure 2.3

Question 3

3. Figure 3.1 shows a 7-segment display driven by a finite state machine (FSM). The FSM output signals A, B, C, D, E, F and G are connected to the display inputs, with the segments turning 'on' when the corresponding signal is asserted. The display is driven such that the sequence 5 $\rightarrow$ 3 $\rightarrow$ 1$\rightarrow$ 0 is repeated. The corresponding patterns on the display are shown in Figure 3.2, where the '1' lies on the right-hand side of the display.

a) Draw the state diagram for the FSM.

[6]

b) Draw the state transition table for the FSM.

[8]

c) The FSM is to be implemented using D-type flip-flops and a combinational logic circuit. Determine the Boolean expressions necessary for this implementation.

[10]

d) Sketch the circuit diagram for your design. This should show the D-type flip-flops, any logic gates, and indicate circuit interconnections.

[6]

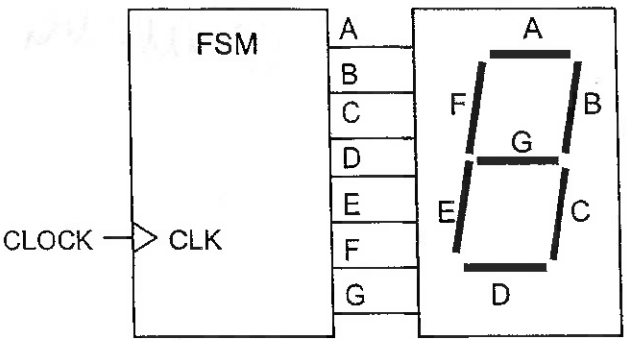


Figure 3.1



Figure 3.2

[THE END]

Seema.

Seat 39. 10.32 query regarding question

10.34 ~~examiner~~ ea requested examiner to come to room

10.40 no announcement

Seat 30 10.42 query (question paper

10.44 for requested examiner to room